## 2nd sessional Thermodynamics- PIPC-201

Q1. State clausius statement of 2nd law of thermodynamics 03

Q1. 5.0 Kg of water at 40°C is mixed with 1 kg of ice at -4°C. The process of mixing is adiabatic and the system is open to atmosphere. Make calculations for the temperature of mixture and the change in entropy of the system. Take specific heat of ice as 2.1 KJ/kgK 6

Q3. Using steam table find the specific volume, enthalpy and entropy of the steam having dryness fraction 0.8 and pressure 15 bar. Also find the value of enthalpy at same pressure when degree of superheat is 50°C. (Use either steam table or mollier diagram). 6